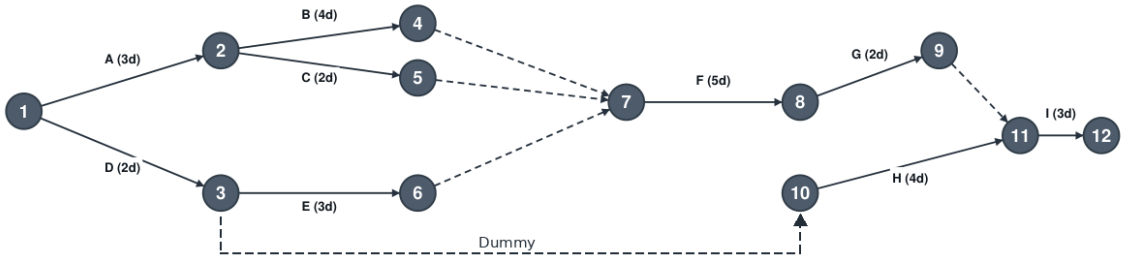
Task 1

GEM MEMBERSHIP AND CLASS BOOKING SYSTEM

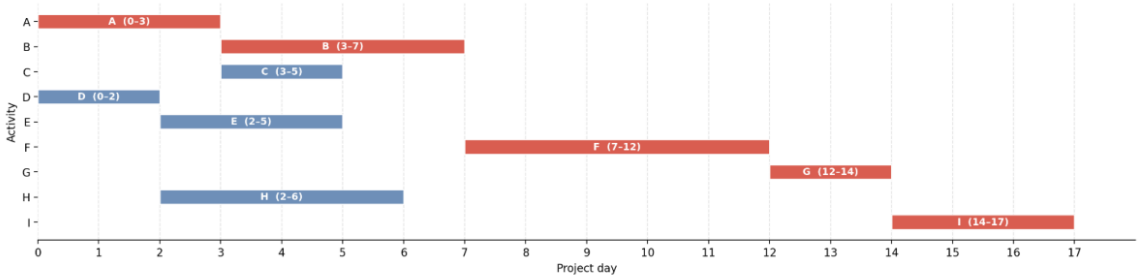
Activities, predecessors, and assumed durations

Activity	Predecessor	Duration (days)
A	—	3
B	A	4
C	A	2
D	—	2
E	D	3
F	B, C, E	5
G	F	2
H	D	4
I	G, H	3

PERT / Precedence Network (duration shown inside each activity)



Gantt Chart | Estimated total project duration: 17 days



Red = critical path | Blue = non-critical

Task 2

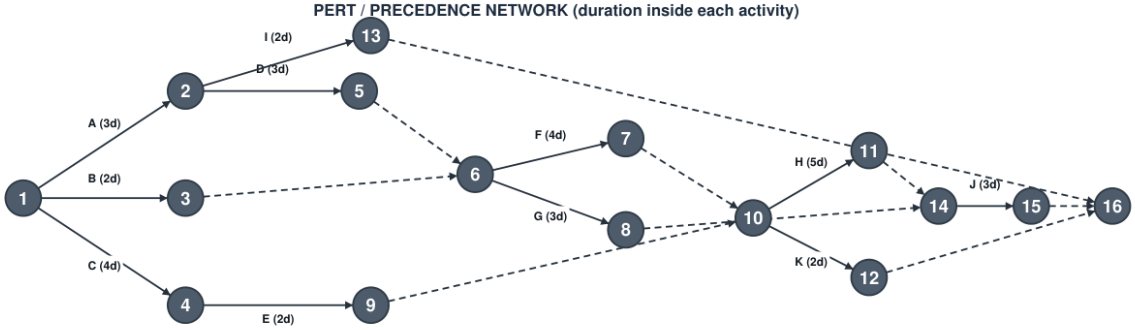
GEM MEMBERSHIP AND CLASS BOOKING SYSTEM

PERT Network and Gantt Schedule

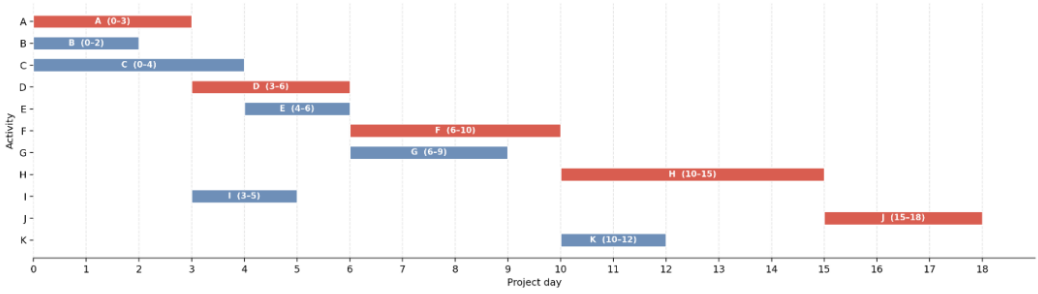
ACTIVITY TABLE

Activity	Predecessor	Duration (days)
A	—	3
B	—	2
C	—	4
D	A	3
E	C	2
F	B, D	4
G	B, D	3
H	E, F	5
I	A	2
J	G, H	3
K	E, F	2

PERT / PRECEDENCE NETWORK (duration inside each activity)



GANTT CHART | ESTIMATED PROJECT DURATION: 18 DAYS



Red = critical path | Blue = non-critical activities

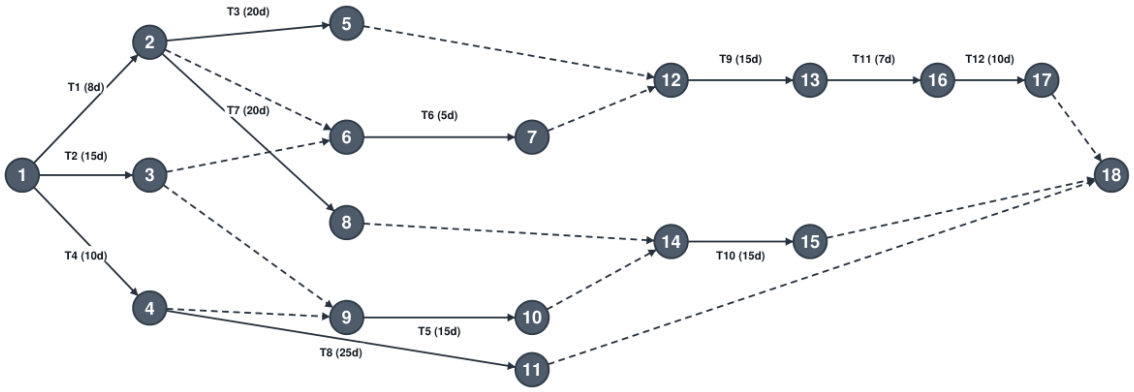
Task 3

GEM MEMBERSHIP AND CLASS BOOKING SYSTEM

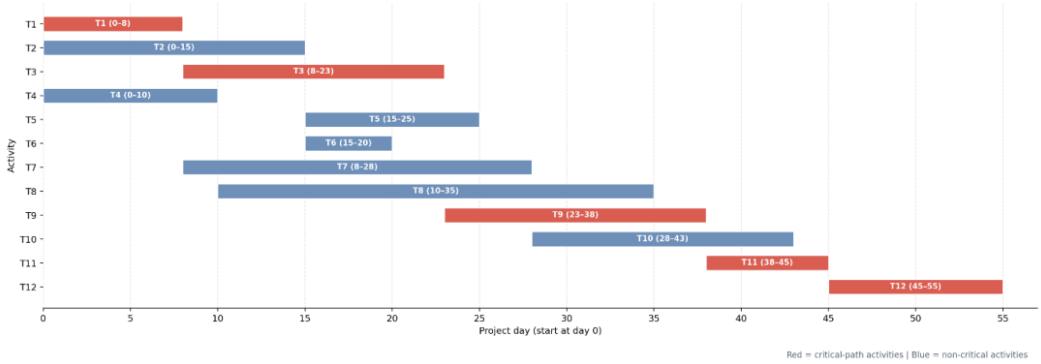
Activity Table — predecessor relationships and durations

Activity	Duration (Days)	Predecessor
T1	8	—
T2	15	—
T3	15	T1
T4	10	—
T5	10	T2, T4
T6	5	T1, T2
T7	20	T1
T8	25	T4
T9	15	T3, T6
T10	15	T5, T7
T11	7	T9
T12	10	T11

PERT / PRECEDENCE NETWORK — duration inside each activity



GANTT CHART | ESTIMATED TOTAL DURATION: 55 DAYS



Red = critical-path activities | Blue = non-critical activities